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Java SE 27 & JDK 27

Class Byte

java.lang.Object
    java.lang.Number
        java.lang.Byte

All Implemented Interfaces:

Serializable, Comparable<Byte>, Constable

```
public final class Byte
extends Number
implements Comparable<Byte>, Constable
```

The `Byte` class is the wrapper class for values of the primitive type `byte`. An object of type `Byte` contains a single field whose type is `byte`.

In addition, this class provides several methods for converting a `byte` to a `String` and a `String` to a `byte`, as well as other constants and methods useful when dealing with a `byte`.

This is a value-based class; programmers should treat instances that are equal as interchangeable and should not use instances for synchronization, or unpredictable behavior may occur. For example, in a future release, synchronization may fail.

Since:

1.1

See Also:

Number, Serialized Form

Field Summary

Fields

Modifier and Type	Field	Description
`static final int`	**BYTES**	The number of bytes used to represent a `byte` value in two's complement binary form.
`static final byte`	**MAX_VALUE**	A constant holding the maximum value a `byte` can have, 2^7-1.
`static final byte`	**MIN_VALUE**	

		A constant holding the minimum value a byte can have, -2^7.
static final int	SIZE	The number of bits used to represent a byte value in two's complement binary form.
static final Class<Byte>	TYPE	The Class instance representing the primitive type byte.

Constructor Summary

Constructors

Constructor	Description
Byte(byte value)	**Deprecated.** It is rarely appropriate to use this constructor.
Byte(String s)	**Deprecated.** It is rarely appropriate to use this constructor.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
byte	byteValue()	Returns the value of this Byte as a byte.
static int	compare(byte x, byte y)	Compares two byte values numerically.
int	compareTo(Byte anotherByte)	Compares two Byte objects numerically.
static int	compareUnsigned(byte x, byte y)	Compares two byte values numerically treating the values as unsigned.

static Byte	**decode**(String nm)	Decodes a String into a Byte.
Optional<DynamicConstantDesc·	**describeConstable**()	Returns an Optional containing the nominal descriptor for this instance.
double	**doubleValue**()	Returns the value of this Byte as a double after a widening primitive conversion.
boolean	**equals**(Object obj)	Compares this object to the specified object.
float	**floatValue**()	Returns the value of this Byte as a float after a widening primitive conversion.
int	**hashCode**()	Returns a hash code for this Byte; equal to the result of invoking intValue().
static int	**hashCode**(byte value)	Returns a hash code for a byte value; compatible with Byte.hashCode().
int	**intValue**()	Returns the value of this Byte as an int after a widening primitive conversion.
long	**longValue**()	Returns the value of this Byte as a long after a widening primitive conversion.
static byte	**parseByte**(String s)	

		Parses the string argument as a signed decimal byte.
static byte	**parseByte**(String s, int radix)	Parses the string argument as a signed byte in the radix specified by the second argument.
short	**shortValue**()	Returns the value of this Byte as a short after a widening primitive conversion.
String	**toString**()	Returns a String object representing this Byte's value.
static String	**toString**(byte b)	Returns a new String object representing the specified byte.
static int	**toUnsignedInt**(byte x)	Converts the argument to an int by an unsigned conversion.
static long	**toUnsignedLong**(byte x)	Converts the argument to a long by an unsigned conversion.
static Byte	**valueOf**(byte b)	Returns a Byte instance representing the specified byte value.
static Byte	**valueOf**(String s)	Returns a Byte object holding the value given by the specified String.
static Byte	**valueOf**(String s, int radix)	Returns a Byte object holding the value extracted from the specified String when parsed with the radix given by the second argument.

Methods declared in class Object

`clone`, `finalize`, `getClass`, `notify`, `notifyAll`, `wait`, `wait`, `wait`

Field Details

MIN_VALUE

```
public static final byte MIN_VALUE
```

A constant holding the minimum value a `byte` can have, -2^7.

See Also:

Constant Field Values

MAX_VALUE

```
public static final byte MAX_VALUE
```

A constant holding the maximum value a `byte` can have, 2^7-1.

See Also:

Constant Field Values

TYPE

```
public static final Class<Byte> TYPE
```

The `Class` instance representing the primitive type `byte`.

SIZE

```
public static final int SIZE
```

The number of bits used to represent a `byte` value in two's complement binary form.

Since:

1.5

See Also:

Constant Field Values

BYTES

```
public static final int BYTES
```

The number of bytes used to represent a `byte` value in two's complement binary form.

Since:

1.8

See Also:

Constant Field Values

Constructor Details

Byte

```
@Deprecated(since="9")
public Byte(byte value)
```

> **Deprecated.**
> *It is rarely appropriate to use this constructor. The static factory `valueOf(byte)` is generally a better choice, as it is likely to yield significantly better space and time performance.*

Constructs a newly allocated `Byte` object that represents the specified `byte` value.

Parameters:

`value` - the value to be represented by the `Byte`.

Byte

```
@Deprecated(since="9")
public Byte(String s)
     throws NumberFormatException
```

> **Deprecated.**
> *It is rarely appropriate to use this constructor. Use `parseByte(String)` to convert a string to a `byte` primitive, or use `valueOf(String)` to convert a string to a `Byte` object.*

Constructs a newly allocated `Byte` object that represents the `byte` value indicated by the `String` parameter. The string is converted to a `byte` value in exactly the manner used by the `parseByte` method for radix 10.

Parameters:

`s` - the `String` to be converted to a `Byte`

Throws:

`NumberFormatException` - if the `String` does not contain a parsable `byte`.

Method Details

toString

`public static String toString(byte b)`

Returns a new `String` object representing the specified `byte`. The radix is assumed to be 10.

Parameters:

`b` - the `byte` to be converted

Returns:

the string representation of the specified `byte`

See Also:

`Integer.toString(int)`

describeConstable

`public Optional<DynamicConstantDesc<Byte>> describeConstable()`

Returns an `Optional` containing the nominal descriptor for this instance.

Specified by:

`describeConstable` in interface `Constable`

Returns:

an `Optional` describing the `Byte` instance

Since:

15

valueOf

`public static Byte valueOf(byte b)`

Returns a `Byte` instance representing the specified `byte` value. If a new `Byte` instance is not required, this method should generally be used in preference to the constructor `Byte(byte)`, as this method is likely to yield significantly better space and time performance since all byte values are cached.

Parameters:

`b` - a byte value.

Returns:

a `Byte` instance representing `b`.

Since:

1.5

parseByte

```
public static byte parseByte(String s,
                             int radix)
                      throws NumberFormatException
```

Parses the string argument as a signed `byte` in the radix specified by the second argument. The characters in the string must all be digits, of the specified radix (as determined by whether `Character.digit(char, int)` returns a nonnegative value) except that the first character may be an ASCII minus sign `'-'` (`'\u002D'`) to indicate a negative value or an ASCII plus sign `'+'` (`'\u002B'`) to indicate a positive value. The resulting `byte` value is returned.

An exception of type `NumberFormatException` is thrown if any of the following situations occurs:

- The first argument is `null` or is a string of length zero.
- The radix is either smaller than `Character.MIN_RADIX` or larger than `Character.MAX_RADIX`.
- Any character of the string is not a digit of the specified radix, except that the first character may be a minus sign `'-'` (`'\u002D'`) or plus sign `'+'` (`'\u002B'`) provided that the string is longer than length 1.
- The value represented by the string is not a value of type `byte`.

Parameters:

`s` - the `String` containing the `byte` representation to be parsed

`radix` - the radix to be used while parsing `s`

Returns:

the `byte` value represented by the string argument in the specified radix

Throws:

`NumberFormatException` - If the string does not contain a parsable `byte`.

parseByte

```
public static byte parseByte(String s)
                      throws NumberFormatException
```

Parses the string argument as a signed decimal `byte`. The characters in the string must all be decimal digits, except that the first character may be an ASCII minus sign `'-'` (`'\u002D'`) to indicate a negative value or an ASCII plus sign `'+'` (`'\u002B'`) to indicate a positive value. The resulting `byte` value is returned, exactly as if the argument and the radix 10 were given as arguments to the `parseByte(java.lang.String, int)` method.

Parameters:

s - a String containing the byte representation to be parsed

Returns:

the byte value represented by the argument in decimal

Throws:

NumberFormatException - if the string does not contain a parsable byte.

valueOf

```
public static Byte valueOf(String s,
                           int radix)
                    throws NumberFormatException
```

Returns a Byte object holding the value extracted from the specified String when parsed with the radix given by the second argument. The first argument is interpreted as representing a signed byte in the radix specified by the second argument, exactly as if the argument were given to the parseByte(java.lang.String, int) method. The result is a Byte object that represents the byte value specified by the string.

In other words, this method returns a Byte object equal to the value of:

```
Byte.valueOf(Byte.parseByte(s, radix))
```

Parameters:

s - the string to be parsed

radix - the radix to be used in interpreting s

Returns:

a Byte object holding the value represented by the string argument in the specified radix.

Throws:

NumberFormatException - If the String does not contain a parsable byte.

valueOf

```
public static Byte valueOf(String s)
                    throws NumberFormatException
```

Returns a Byte object holding the value given by the specified String. The argument is interpreted as representing a signed decimal byte, exactly as if the argument were given to the parseByte(java.lang.String) method. The result is a Byte object that represents the byte value specified by the string.

In other words, this method returns a Byte object equal to the value of:

```
Byte.valueOf(Byte.parseByte(s))
```

Parameters:

`s` - the string to be parsed

Returns:

a `Byte` object holding the value represented by the string argument

Throws:

`NumberFormatException` - If the `String` does not contain a parsable `byte`.

decode

```
public static Byte decode(String nm)
                   throws NumberFormatException
```

Decodes a `String` into a `Byte`. Accepts decimal, hexadecimal, and octal numbers given by the following grammar:

DecodableString:
- $Sign_{opt}$ *DecimalNumeral*
- $Sign_{opt}$ `0x` *HexDigits*
- $Sign_{opt}$ `0X` *HexDigits*
- $Sign_{opt}$ `#` *HexDigits*
- $Sign_{opt}$ `0` *OctalDigits*

Sign:
- `-`
- `+`

DecimalNumeral, *HexDigits*, and *OctalDigits* are as defined in section §3.10.1 of *The Java Language Specification*, except that underscores are not accepted between digits.

The sequence of characters following an optional sign and/or radix specifier ("`0x`", "`0X`", "`#`", or leading zero) is parsed as by the `Byte.parseByte` method with the indicated radix (10, 16, or 8). This sequence of characters must represent a positive value or a `NumberFormatException` will be thrown. The result is negated if first character of the specified `String` is the minus sign. No whitespace characters are permitted in the `String`.

Parameters:

`nm` - the `String` to decode.

Returns:

a `Byte` object holding the `byte` value represented by `nm`

Throws:

`NumberFormatException` - if the `String` does not contain a parsable `byte`.

See Also:

`parseByte(java.lang.String, int)`

byteValue

```
public byte byteValue()
```

Returns the value of this `Byte` as a `byte`.

Overrides:

`byteValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type `byte`.

shortValue

```
public short shortValue()
```

Returns the value of this `Byte` as a `short` after a widening primitive conversion.

Overrides:

`shortValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type `short`.

See *Java Language Specification*:

5.1.2 Widening Primitive Conversion

intValue

```
public int intValue()
```

Returns the value of this `Byte` as an `int` after a widening primitive conversion.

Specified by:

`intValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type `int`.

See *Java Language Specification*:

5.1.2 Widening Primitive Conversion

longValue

```
public long longValue()
```

Returns the value of this `Byte` as a `long` after a widening primitive conversion.

Specified by:

`longValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type `long`.

See *Java Language Specification*:

5.1.2 Widening Primitive Conversion

floatValue

```
public float floatValue()
```

Returns the value of this `Byte` as a `float` after a widening primitive conversion.

Specified by:

`floatValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type `float`.

See *Java Language Specification*:

5.1.2 Widening Primitive Conversion

doubleValue

```
public double doubleValue()
```

Returns the value of this `Byte` as a `double` after a widening primitive conversion.

Specified by:

`doubleValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type `double`.

See *Java Language Specification*:

5.1.2 Widening Primitive Conversion

toString

```
public String toString()
```

Returns a `String` object representing this `Byte`'s value. The value is converted to signed decimal representation and returned as a string, exactly as if the `byte` value were given as an argument to the `toString(byte)` method.

Overrides:

`toString` in class `Object`

Returns:

a string representation of the value of this object in base 10.

hashCode

```
public int hashCode()
```

Returns a hash code for this `Byte`; equal to the result of invoking `intValue()`.

Overrides:

`hashCode` in class `Object`

Returns:

a hash code value for this `Byte`

See Also:

`Object.equals(java.lang.Object)`,
`System.identityHashCode(Object)`

hashCode

```
public static int hashCode(byte value)
```

Returns a hash code for a `byte` value; compatible with `Byte.hashCode()`.

Parameters:

`value` - the value to hash

Returns:

a hash code value for a `byte` value.

Since:

1.8

equals

```
public boolean equals(Object obj)
```

Compares this object to the specified object. The result is `true` if and only if the argument is not `null` and is a `Byte` object that contains the same `byte` value as this object.

Overrides:

`equals` in class `Object`

Parameters:

`obj` - the object to compare with

Returns:

`true` if the objects are the same; `false` otherwise.

See Also:

`Object.hashCode()`, `HashMap`

compareTo

```
public int compareTo(Byte anotherByte)
```

Compares two `Byte` objects numerically.

Specified by:

`compareTo` in interface `Comparable<Byte>`

Parameters:

`anotherByte` - the `Byte` to be compared.

Returns:

the value `0` if this `Byte` is equal to the argument `Byte`; a value less than `0` if this `Byte` is numerically less than the argument `Byte`; and a value greater than `0` if this `Byte` is numerically greater than the argument `Byte` (signed comparison).

Since:

1.2

compare

```
public static int compare(byte x,
                          byte y)
```

Compares two `byte` values numerically. The value returned is identical to what would be returned by:

```
Byte.valueOf(x).compareTo(Byte.valueOf(y))
```

Parameters:

`x` - the first `byte` to compare

`y` - the second `byte` to compare

Returns:

the value `0` if `x == y`; a value less than `0` if `x < y`; and a value greater than `0` if `x > y`

Since:

1.7

compareUnsigned

```
public static int compareUnsigned(byte x,
                                  byte y)
```

Compares two `byte` values numerically treating the values as unsigned.

Parameters:

`x` - the first `byte` to compare

`y` - the second `byte` to compare

Returns:

the value `0` if `x == y`; a value less than `0` if `x < y` as unsigned values; and a value greater than `0` if `x > y` as unsigned values

Since:

9

toUnsignedInt

```
public static int toUnsignedInt(byte x)
```

Converts the argument to an `int` by an unsigned conversion. In an unsigned conversion to an `int`, the high-order 24 bits of the `int` are zero and the low-order 8 bits are equal to the bits of the `byte` argument. Consequently, zero and positive `byte` values are mapped to a numerically equal `int` value and negative `byte` values are mapped to an `int` value equal to the input plus 2^8.

Parameters:

`x` - the value to convert to an unsigned `int`

Returns:

the argument converted to `int` by an unsigned conversion

Since:

1.8

toUnsignedLong

```
public static long toUnsignedLong(byte x)
```

Converts the argument to a `long` by an unsigned conversion. In an unsigned conversion to a `long`, the high-order 56 bits of the `long` are zero and the low-order 8 bits are equal to the bits of the `byte` argument. Consequently, zero and positive `byte` values are mapped to a numerically equal `long` value and negative `byte` values are mapped to a `long` value equal to the input plus 2^8.

Parameters:

`x` - the value to convert to an unsigned `long`

Returns:

the argument converted to `long` by an unsigned conversion

Since:

1.8

Report a bug or suggest an enhancement
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